

Search for events in XENON1T associated with Gravitational Waves

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We perform a blind search for particle signals in the XENON1T dark matter detector that occur close in time to gravitational wave signals in the LIGO and Virgo observatories. No particle signal

is observed in the nuclear recoil ~~, electronic recoil, CE ν NS, and S2-only and electronic recoil~~ channels within ± 500 seconds of observations of the gravitational wave signals GW170104, GW170729, GW170817, GW170818, and GW170823. We use this null result to constrain mono-energetic neutrinos and Beyond Standard Model particles emitted in the closest coalescence GW170817, a binary neutron star merger. We set new upper limits on the fluence (time-integrated flux) of coincident neutrinos down to 17 keV at 90% confidence level. Furthermore, we constrain the product of coincident fluence and cross section of ~~Beyond Standard Model particles~~ axion-like particles (ALPs) to be less than 10^{-29} cm²/cm² in the [5.5–210] keV energy range at 90% confidence level.

Keywords: Dark Matter, Direct Detection, Xenon

I. INTRODUCTION

Following the first detection of a gravitational wave (GW) signal by the Laser Interferometer Gravitational-Wave Observatory (LIGO) in 2016 [1], GW signals from binary mergers are now routinely detected by the LIGO and Virgo observatories [2]. The subsequent observation of electromagnetic counterparts from binary neutron star coalescence in 2017 [3]–[4] has made GW signals an integral part of multi-messenger astrophysics. Various detectors have conducted follow-up searches for neutrinos associated with binary mergers in a time window of $\pm [400\text{--}500]$ s of the observed GW signal [5?–10] [5–12], which might indicate the simultaneous emission of relativistic particles. The sensitive (neutrino) energy ranges of these searches, summarized in Tab. I, span from EeV down to half an MeV. No evidence of a particle signal, coincident with GW signal, has been reported yet. ~~Here, we use data from the XENON1T dark matter search [13] to extend this neutrino energy window down to 17 keV. Further, the low threshold of 1 keV electron-equivalent (keV_{ee}) offered by the XENON1T, enables us to search for Beyond Standard Model (BSM) particles from the merger of neutron stars, such as relativistic axions [14, 15] or sterile neutrinos [16, 17].~~

TABLE I. Search for particles coincident with GW and the energy ranges they are sensitive to.

Detector	Detection energy
Pierre Auger Observatory <u>Observatory</u> [5]	10^{11} MeV <u>MeV</u>
IceCube [6]	10^5 MeV <u>MeV</u>
<u>ANTARES [11]</u>	10^5 MeV <u>MeV</u>
Super-K [7]	3.5 MeV <u>MeV</u>
XMASS [8]	14 MeV <u>MeV</u>
KamLAND [9]	1.8 MeV <u>MeV</u>
NOvA [2]–[12]	10 MeV <u>MeV</u>
Borexino [10]	0.5 MeV <u>MeV</u>
XENON1T (this work)	0.017 MeV <u>MeV</u>

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Here, we use data from the XENON1T dark matter search [13] to extend this neutrino energy window down to 17 keV. Further, the low threshold of 1 keV electron-equivalent (keV_{ee}) offered by the XENON1T, enables us to search for Beyond Standard Model (BSM) particles in the keV energy range. There have been studies proposing the emission of relativistic axions [14, 15] as well as sterile neutrinos [16, 17] from the merger of neutron stars. Since blackbody radiation peaks at several to hundreds of keV for binary neutron stars [18], BSM particles might exist in thermal equilibrium with the blackbody radiation with keV energy. Additionally, BSM particles can be accelerated to keV energies by 511-keV line associated with positrons produced by mergers [19]. Not only the low energy thresholds of XENON1T are helpful in looking for these keV particles but they also allow looking for MeV BSM particles which deposit only a fraction of their energy via interactions like coherent scattering [20].

II. DATA SETS

The LIGO and Virgo observatories conducted their second observation run from November 30, 2016, to August 25, 2017 [26]. Overlapping with this time frame, the XENON1T detector took Science Run 0 (SR0) data from November 22, 2016 to January 18, 2017, and Science Run 1 (SR1) data from February 2, 2017 to February 8, 2018. This data has already been analyzed in several ~~channels~~ modes that we revisit here in order to check for particle signals coincident with LIGO and Virgo GW observations.

The XENON1T experiment has been described in [13]. ~~A search for nuclear recoils~~ There are two channels via which a particle can interact with the detector. A particle can scatter off the xenon nuclei, referred to as nuclear recoil (NR). Similarly, a particle can scatter off the electrons of xenon atoms, referred to as electronic recoil (ER) [23]. There are three ways in which searches for NR signals have been performed. The search for weakly interacting massive particles has been done demanding both scintillation light (S1) and electroluminescence from ionization electrons (S2) is available during both SR0 [21] and SR1 [22]. In addition, using data from SR1, a search was conducted for electronic recoils (ER) [23], for nuclear recoils [21, 22]. This will be referred to as 3-fold in the rest of the paper because coincident signals from at

TABLE II. Characteristics of ~~analysis channels~~ analyses considered in this study. For each of these ~~channels~~ analyses, we show the corresponding energy ranges, the Science Runs that have been analyzed for that channel, the fiducial target mass, and the observed background rate. Further, in the ± 500 s time window around a GW event, we give the average livetime for detecting a signal, the average number of expected background events (N_{exp}), and the probability of observing zero background events ($P(0)$).

	Event-							
	Energy		Runs	Fiducial	Background rate	Livetime		
channel <u>Analyses</u>	range		analyzed	target mass	(events/day)	(per GW)	N_{exp}	$P(0)$
NR [21, 22] <u>ER [23]</u>	4.91 keV _{nr}	40.9210 keV _{nr}	SR0, SR1	1.31.04 tonne	0.05-186	935 s	0.00054 <u>See Tab. IV</u>	0.9995 <u>See Tab. IV</u>
ER [23] <u>3-fold [21, 22]</u>	14.9 keV _{ee}	21040.9 keV _{ee}	SR0, SR1	1.041.3 tonne	186-0.05	935 s	See Tab. IV <u>0.00054</u>	See Tab. IV <u>0.9995</u>
CEνNS 2-fold [24]	≥ 1.6 keV _{nr}	≥ 1.6 keV _{nr}	SR1	1.04 tonne	0.03	790 s	0.00027	0.9997
S2-only [25]	≥ 0.7 keV _{nr}	≥ 0.7 keV _{nr}	SR1	0.12 tonne	0.94	650 s	0.0071	0.993

least three photomultiplier tubes (PMTs) are required for an S1. However, with the purpose of decreasing the energy threshold, the data have been analyzed in two other modes. To detect NRs produced by solar neutrinos via coherent elastic neutrino-nucleus scattering (CE ν NS) [24], and for nuclear recoils where, the requirement for an S1 has been loosened to coincident signals from at least two PMTs [24]. This will be referred to as 2-fold in the rest of the paper. To detect NRs produced by light dark matter, the requirement of a scintillation signal has been waived [25], referred to as S2-only [25] in the rest of the paper.

The 3-fold search for NRs was done using data from both SR0 and SR1 data. However, the search for ERs or 2-fold and S2-only searches for NRs were done only using data from SR1. Details of these ~~analysis channels~~ analyses can be found in the respective papers [21–25]. ~~Notably, the NR and ER channels~~ To re-iterate, the search for ERs and the 3-fold search for NRs demand coincident signals from at least three ~~photomultiplier tubes (PMTs)~~ PMTs for an S1, while the ~~CE ν NS channel 2-fold~~ requires signals from only two PMTs, and ~~S2-only doesn't require S1~~. Furthermore, the S2 threshold has been reduced from 200 photoelectrons (PE) ~~and 500 PE for the NR and ER channels, respectively, for 3-fold analysis~~ to 120 PE and 150 PE for ~~the CE ν NS 2-fold and S2-only channels~~ analyses, respectively.

The S2 threshold for ER channel is 500 PE. The corresponding energy ranges of these ~~channels~~ analyses are given in Tab. II in units of keV electron-equivalent (keV_{ee}) for ~~the ER channel~~, and as keV nuclear recoil-equivalent (keV_{nr}) for ~~the NR, CE ν NS, and NR channel~~.

The lower threshold for 2-fold and S2-only analyses comes with the cost of increased backgrounds, to reduce which, stringent cuts need to be applied. For 2-fold analysis, these cuts result in low total acceptance – peak acceptance is 15%. The acceptance is lower than 10% for energy region outside 2–4 keV_{nr}, becoming negligible above 6 keV_{nr}; compared to $\sim 80\%$ acceptance for 3-fold

and S2-only ~~channels~~. Effective xenon target masses vary for different analyses. Whereas the exposure of the NR, ER, and CE ν NS channels is limited by the fiducial volume, the analyses within 2–20 keV_{nr} and 10–40 keV_{nr} respectively. On the other hand, the stringent radius and width cuts for S2-only channel is limited by the strong event selections applied to reduce backgrounds analysis lead to the fiducial target mass of 0.12 tonne compared to fiducial target mass of > 1 tonne in 2-fold and 3-fold analyses.

The background rate is very low for ~~the NR, CE ν NS, and S2-only channels~~ NR channel (see Tab. II), which motivates ~~the use of using~~ a single energy bin for each of ~~these channels~~ analysis mode. However, the background rate is quite high in ER channel due to the β decay of ^{214}Pb , present due to ^{222}Rn emanation by materials [23]. Further, from Fig. 3 of [23], we can see that the ER spectrum is not constant in energy. The spectrum exhibits spectral lines from radioactive $^{83\text{m}}\text{Kr}$ and $^{131\text{m}}\text{Xe}$. We thus define five energy bins for the ER channel: low energy (0–30 keV_{ee}), $^{83\text{m}}\text{Kr}$ (30–50 keV_{ee}), medium energy (50–142 keV_{ee}), $^{131\text{m}}\text{Xe}$ (142–185 keV_{ee}) and high energy (185–210 keV_{ee}).

Three of the eight GW signals that LIGO/VIRGO observed during SR0 and SR1 occurred during XENON1T calibration or downtime. ~~Only SR0 NR data is available during GW170104, while SR1 data in all four analysis channels is available during~~ was during SR0 and therefore only 3-fold analysis is available for it. GW170729, GW170817, GW170818, and GW170823 were during SR1, and so all four analyses are available for them. Out of these GWs, GW170817 was due to the merger of neutron stars ~~, and the only GW event with a co-detection of electromagnetic signatures.~~ Further, it was at a distance of 40 Mpc, compared to a distance of ~ 1000 Mpc for other events. ~~Out of these GWs, While the particle flux is dominated by GW170817 on the Earth would be influenced by initial masses, nuclear equation of state, and anisotropies of emission [27], the proximity to merger also plays a role~~ because the flux is inversely pro-

portional to the square of ~~distance~~. the distance. Because of all these reasons, GW170817 is the most interesting and would be used to derive our results in section V.

III. NEUTRINO SPECTRA

Neutrinos predominantly interact in our detector via elastic scattering off electrons or xenon nuclei. The standard model differential cross section for elastic neutrino-electron scattering can be written as [28, 29],

$$\frac{d\sigma_{\text{ES}}}{dE_r} = \frac{G_F^2 m_e}{2\pi} \left[(g_V + g_A)^2 - (g_V^2 - g_A^2) \frac{m_e E_r}{E_\nu^2} + (g_V - g_A)^2 \left(1 - \frac{E_r}{E_\nu} \right)^2 \right],$$

$$\frac{d\sigma_{\text{ES}}}{dE_r} = \frac{G_F^2 m_e}{2\pi} \left[(g_V + g_A)^2 - (g_V^2 - g_A^2) \frac{m_e E_r}{E_\nu^2} + (g_V - g_A)^2 \left(1 - \frac{E_r}{E_\nu} \right)^2 \right], \quad (1)$$

where E_r is the ~~electron recoil-ER~~ kinetic energy, E_ν is the incoming neutrino energy, m_e is the mass of an electron, and G_F is the Fermi constant. Neutrino-electron coupling constants for different flavors of neutrinos and antineutrinos are furnished in Tab. III.

TABLE III. Neutrino-electron coupling constants for different flavors of neutrinos and antineutrinos. θ_W is the Weinberg angle.

Neutrino flavor	g_V	g_A
ν_e	$2 \sin^2 \theta_W + \frac{1}{2}$	$+\frac{1}{2}$
$\bar{\nu}_e$	$2 \sin^2 \theta_W + \frac{1}{2}$	$-\frac{1}{2}$
ν_μ, ν_τ	$2 \sin^2 \theta_W - \frac{1}{2}$	$-\frac{1}{2}$
$\bar{\nu}_\mu, \bar{\nu}_\tau$	$2 \sin^2 \theta_W - \frac{1}{2}$	$+\frac{1}{2}$

The expected number of signals in the ER channel is given as,

$$\frac{dN_{\text{ER}}}{dE_r} = N_T(E_r) \epsilon(E_r) \int \frac{d\sigma_{\text{ES}}}{dE_r} \frac{df(E_\nu)}{dE_\nu} dE_\nu, \quad (2)$$

where $\epsilon(E_r)$ is the detection efficiency of the ER channel as a function of the recoil energy [23]. $f(E_\nu)$ is the neutrino energy spectrum. ~~Given an absence of theoretical predictions, we assume that neutrinos are emitted mono-energetically in binary mergers, as done in previous low-energy searches [7, 8, 10].~~ $N_T(E_r)$ is

the total number of available electrons in the fiducial target: $N_T(E_r) = N_{\text{Xe}} \times N_e$, where N_{Xe} is the total number of xenon atoms in the fiducial volume and N_e is number of electrons per xenon atom with binding energies below the particular recoil energy. ~~We use the free electron approximation because a more precise relativistic random phase approximation is unavailable for our energy region of interest.~~ The different shells of a xenon atom, along with the number of electrons and the potential energy, are given in [30]. ~~We also convolve the cross section with a Gaussian function with standard deviation $0.31\sqrt{E}\sqrt{\text{keV}} + 0.0037E$ [23] to account for energy resolution, where E is in keV.~~

The effective cross section as a function of neutrino energy, taking into account the detector properties, is shown in Fig. 1 (left). ~~We use the free electron approximation because a more precise relativistic random phase approximation is unavailable for our energy region of interest. We convolve the cross section with a Gaussian function with standard deviation $0.31\sqrt{E}\sqrt{\text{keV}} + 0.0037E$ [23] to account for energy resolution, where E is in keV. We also account for the fact that~~ As can be seen in equation 1, neutrino energy is a higher-order term for differential cross section. The change in the physical cross section of electron neutrinos, electron antineutrinos, muon or tau neutrinos, and muon or tau antineutrinos with neutrino energy is only 16%, 54%, 1.2%, and 5% respectively. However, to satisfy the kinematics of the system, a neutrino can not produce a recoil of energy greater than $2E_\nu^2/(2E_\nu + m_e)$ [29]. ~~Our threshold of 1 keV_{ee} makes us insensitive to neutrinos of energy 16 keV or below. Given an absence of theoretical predictions, we assume that neutrinos are emitted mono-energetically in binary mergers, as done in previous low-energy searches [7, 8, 10]. The cross section for electronic recoils depends weakly on neutrino energy; it is almost constant at neutrino energies above 36 keV. However, it decreases below 36 keV because low-energy~~ i.e., the cross-section sharply goes to 0 at this recoil energy for a given neutrino energy. Our upper threshold for ER energy is 210 keV_{ee}, which is the maximum ER energy produced by a 360 keV neutrino. Lower energy neutrinos are incapable of generating ~~electronic recoils~~ ERs throughout the entire energy range of the ER channel (1 keV_{ee} – 210 keV_{ee}). This leads to a drastic decrease in the effective cross section below 360 keV. Further, our lower threshold of 1 keV_{ee} makes us insensitive to neutrinos of energy 16 keV or below.

Having discussed the interaction of neutrinos with electrons, we now turn to the interaction of neutrinos with xenon nuclei. Neutrinos in the energy range of 10 – 50 MeV can interact coherently with nuclei [20]. For CE ν NS, we use the same differential cross section as in [24]. The ~~number of nuclear recoils in the detector~~ expected number of signals in the NR channel has a similar form as Eq. (2), with $N_T(E_r)$ and σ_{ES} replaced by N_{Xe} and $\sigma_{\text{CE}\nu\text{NS}}$ respectively. Also, $\epsilon(E_r)$ is the detection efficiency of considered ~~channels-analysis mode~~

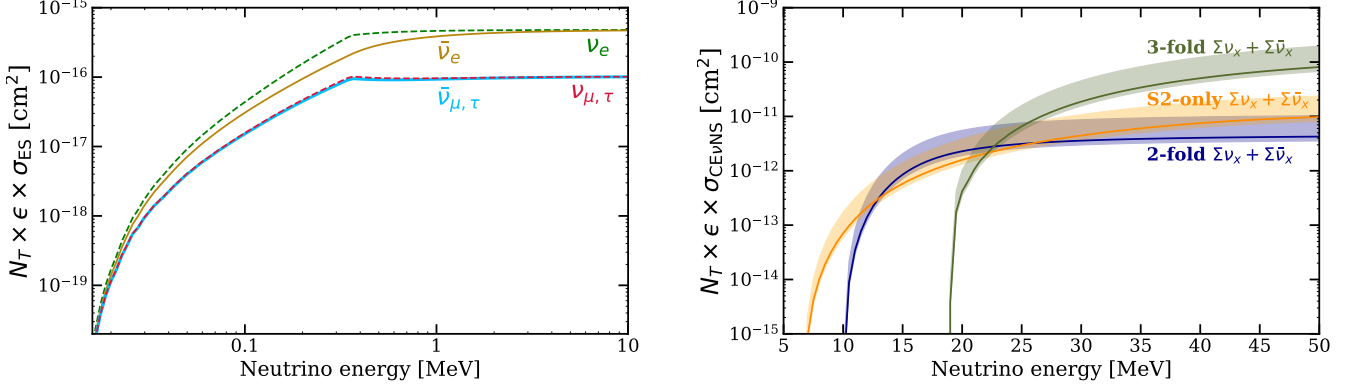


FIG. 1. Left: the effective cross section for elastic neutrino-electron scattering as a function of neutrino energy for electron neutrinos (ν_e , dashed green), electron antineutrinos ($\bar{\nu}_e$, solid gold), muon or tau neutrinos ($\nu_{\mu,\tau}$, dashed red), and muon or tau antineutrinos ($\bar{\nu}_{\mu,\tau}$, solid cyan). Right: the effective cross section for CE ν NS as a function of neutrino energy. NR-3-fold (olive-green), CE ν NS-2-fold (blue), and S2-only (orange) refer to different ways modes in which the data was analyzed in the search for nuclear recoils, as described in Sec. II. In the right plot, the cross section is for the interaction of all flavors of neutrinos and antineutrinos.

as a function of the nuclear-recoil-NR kinetic energy. Assuming mono-energetic emission of the neutrinos, the effective cross section as a function of neutrino energy, taking into account the detector properties, is shown in Fig. 1 (right). Effective cross section is highest for 3-fold analysis because of larger acceptance and fiducial mass. The lower fiducial mass for S2-only analysis leads to an order of magnitude lower effective cross section at higher neutrino energy. The very low total acceptance of 2-fold analysis leads to even lower effective cross section. However, because of comparatively higher acceptance at lower recoil energies, the 2-fold effective cross section is comparable or even larger than S2-only effective cross section at lower neutrino energies.

We also consider non-standard neutrino-neutrino-nuclei interactions via the vector coupling of ν_e to up and down quarks. The coupling values within 90% confidence interval from XENON1T data [24], can effectively change the total interaction cross section from a factor of 0.83 to 2.42, under the assumption that $1/6^{th}$ of the total neutrinos are electron neutrinos. The resulting changes in the number of nuclear-recoils-NRs are shown as an uncertainty band in Fig. 1 (right).

IV. BACKGROUND ESTIMATION

Consistent with previous coincidence searches shown in Tab. I, which used a time window of $\pm [400-500]$ s, we use a ± 500 s time window centered on the time of GW signal. This is appropriate for any relativistic particle flux emitted by the binary merger. Accounting for deadtime due to various cuts [31]. However, in XENON1T, various cuts in different analyses remove events at certain times; accounted as deadtime [32]. There are veto cuts to remove events if the data acquisition system is full

and unable to record the data or if there are muons or muon-induced showers passing through the detector. Further, there are cuts to remove lone S1s and S2s produced by energy deposition in light- and background events, livetimes-charge-insensitive regions. These cuts are tightened for 2-fold analysis because of the higher background rate, resulting in its lower livetime. The deadtime due to these cuts is uniformly distributed within the time window. Further, the deadtime due to backgrounds in the detector is [13-14] s, which again is uniformly distributed within the time window. We multiply [986-987] s, with the cut-acceptance of aforementioned cuts to account for deadtime due to the cuts. The livetimes obtained for detecting a signal in the ± 500 s time window around a GW signal in the NR, ER, CE ν NS-ER, 3-fold, 2-fold and S2-only channels are approximately 935 analyses are [926-940] s, 935[926-942] s, 799[785-792] s and 659[920-933] s respectively. For S2-only analysis, 30% of data, uniformly distributed over time, was taken to train the cuts. Therefore, the livetime of S2-only analysis further reduces to [644-653] s. The average livetimes for different analyses are given in Tab II.

Although background events in these channels have been studied before, a direct comparison with the timing of gravitational waves has never been made. To ensure an unbiased analysis, we follow a blind approach where we refrain from utilizing the timing information of individual background events until the background model is finalized and the expected number of background events in the ± 500 s time window is determined.

Fig. 2 shows the distribution of background events with time in different analysis channels. Rates are constant within statistical uncertainty for the duration of the Science Runs for the NR, CE ν NS, and S2-only channels. Therefore, for these channels NR

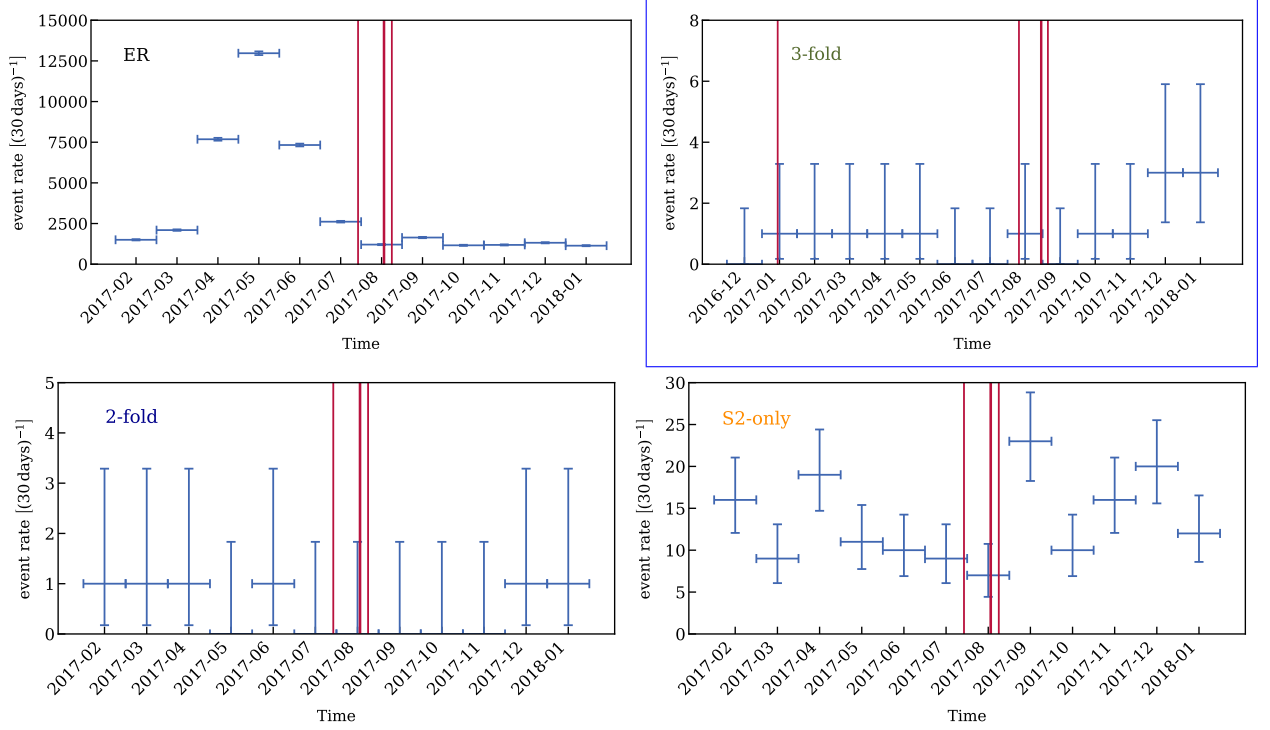


FIG. 2. The distribution of background events with time in [ER and NR channels](#). [3-fold](#), [2-fold](#), and [S2-only](#) refer to different analysis [channels/modes of the NR channel](#). Red lines indicate the time of GW events. The rise and subsequent decay of event rate in the ER channel during March-September are due to production of $^{131\text{m}}\text{Xe}$, ^{133}Xe , and ^{125}I during neutron calibration campaigns [33].

TABLE IV. Predictions for the different energy bins of the ER channel during four GW events. For each bin, in the ± 500 s time window around a GW event, we give the number of expected background events (N_{exp}), the probability of observing zero background events ($P(0)$) and the [upper limits \(UL\) on the expected minimum](#) number of [background events corresponding to an excess](#) at a 3σ confidence level (N_{th}).

	Low Energy			$^{83\text{m}}\text{Kr}$			Medium Energy			$^{131\text{m}}\text{Xe}$			High Energy		
	N_{exp}	$P(0)$	N_{th}	N_{exp}	$P(0)$	N_{th}	N_{exp}	$P(0)$	N_{th}	N_{exp}	$P(0)$	N_{th}	N_{exp}	$P(0)$	N_{th}
GW170729	0.04	0.94	2	0.16	0.85	3	0.22	0.80	3	0.64	0.52	5	0.07	0.93	2
GW170817	0.04	0.94	2	0.15	0.86	3	0.22	0.80	3	0.29	0.75	4	0.07	0.93	2
GW170818	0.04	0.94	2	0.15	0.86	3	0.22	0.80	3	0.27	0.76	4	0.07	0.93	2
GW170823	0.04	0.94	2	0.14	0.87	3	0.22	0.80	3	0.23	0.79	3	0.07	0.93	2

[channel](#), the background rate is the ratio of the total number of background events during the entire Science Run and the total livetime of the entire Science Run.

However, due to the background of activated isotopes — $^{131\text{m}}\text{Xe}$, ^{133}Xe , and ^{125}I ; we need to account for the time-dependence of background rate [in ER channel](#). Thus, we use the same background estimation for the ER channel as in [33]. Before unblinding the timing information of background events, we compare the values predicted by these background models with the actual observations in the different energy bins. The comparison was done using the $(-1500, -500)$ s sideband before the GW events. To account for the degree of freedom due to five independent ER energy bins and four GW signals, we divided the chi-square by 20 and found the reduced

chi-square statistic to be 1.14. This confirms that these background models agree with the observation.

Table II gives background rate and livetime in the units of events/day and seconds respectively. Therefore, the expected number of background events in the ± 500 s time window around a GW signal, is calculated as,

$$N_{\text{exp}} = \text{background rate} \times \frac{\text{livetime}}{24 \times 60 \times 60}. \quad (3)$$

The corresponding probabilities of observing zero background events are values of cumulative distribution functions at zero, assuming Poisson statistics. These values are given in Tab. II for the NR [, CEvNS, and S2-only channels](#) and in Tab. IV for the ER channel.

The ± 500 s time window was blinded during background estimation, and we ~~calculated the upper limit on the expected number of background events in calculate the minimum number of events in each bin of the ER channel, based on the background model for each GW, above which evidence for a particle signal in that bin could have been claimed. For Poisson statistics, the relationship between local~~ using the expected number of background events, based on the background model. A potential excess at a 3σ confidence level corresponds to a p-value ~~and global~~ of 3×10^{-3} . However, this local p-value ~~taking into account needs to be corrected to account for the look-elsewhere effect due to 20 degrees of freedom (n_{dof}) is given by Sidak Correction independent bins. The look-elsewhere effect refers to an increased chance of a false positive for a larger number of trials due to statistical fluctuations [34]. To correct the conclusions of a hypothesis test, Sidak correction defines the global p-value for n independent bins as [35],~~

$$p_{\text{global}} = 1 - (1 - p_{\text{local}})^{1/n_{\text{dof}} 1/n}. \quad (4)$$

The ~~local p-value of 3×10^{-3} corresponds to correction yields~~ a global p-value of 1.5×10^{-4} for a potential excess at a 3σ confidence level. The ~~resulting upper limits on corresponding minimum numbers of required events for reporting excess, calculated using the expected number of background events following Poisson distribution, are between 2 and 5, as shown in Tab. IV. We do not give upper limits on the expected number of background events in the other channels because In the NR channel, values of cumulative distribution functions at 1, for Poisson distributions with such small means, are more than 99.99%; implying that even 1 event corresponds to an excess.~~

V. RESULTS AND CONCLUSIONS

Once these detection channels, time and energy ranges, and background models were fixed, the timing information of background events was unblinded. No background event was observed in any of these channels within ± 500 seconds of any tested GW signal. Given the low background rates, this is expected under the background-only hypothesis in the NR, ~~CEvNS, and S2-only channels~~ channel, as can be seen in Tab II. The total number of expected background events in all energy bins of the ER channel across all GWs is ~ 3 . However, even in this channel, no background events were observed, consistent with the background-only hypothesis, assuming Poisson statistics, within a 3σ confidence interval.

The absence of background events can be converted into an upper limit on coincident neutrinos emitted in the mergers, assuming a Poisson distribution of neutrino signal events [36]. To ensure consistency with other experiments, we place limits at a 90% confidence level on

fluence, which represents the integration of flux with respect to time. ~~We are most sensitive to As explained in section II, these limits are on the fluence from the merger of neutron stars – GW170817, due to its close proximity [3].~~ The upper limit on neutrino fluence at 90% confidence level using ~~electronic recoils ER channel~~ is shown in Fig. 3. As can be seen, we place upper limits on the fluence of $\mathcal{O}(10 \text{ keV})$ neutrinos. No other detector has been able to probe such low-energy neutrinos. The upper limit on neutrino fluence at 90% confidence level using ~~nuclear recoils NR channel~~ is shown in Fig. 4. Our limits are comparable to those from the XMASS dark matter detector. Both limits are on the sum of the fluence of all flavors of neutrinos and antineutrinos, which is different for the stronger limits from Super-K. ~~Further, all the limits (Fig. 3 and 4) correspond to neutrino fluxes higher than the typical mass of the binary systems for isotropic emission.~~

We also use our data to constrain the fluence of ~~BSM particles axion-like particles (ALPs)~~ with keV energies coincident with GW events, ~~such as axions or sterile neutrinos. Similar to neutrinos, we assume that these BSM particles are emitted mono-energetically. Further, in the absence of a concrete model for the interaction of these BSM particles with electrons, we assume that all of their.~~ In XENON1T, ALPs can be detected via the axio-electric effect, a process analogous to the photoelectric effect [37]. Absorption of an axion causes atomic ionization, and the energy is transferred ~~from the axion to the electron during an interaction. For this assumption to be valid, the particles' rest mass must be negligible compared to their total energy of $\mathcal{O}(10 \text{ keV})$. These assumptions yield the interaction. Similar to neutrinos, we assume that ALPs are emitted mono-energetically from mergers. Therefore the number of electronic recoils ERs in the detector due to BSM particles as, ALPs can be expressed as,~~

$$N_{\text{ER}} = N_T \epsilon \sigma_{\text{BSM ALP}} F_{\text{EBSM EALP}}, \quad (5)$$

where ~~$\sigma_{\text{BSM ALP}}$~~ is the cross section of ~~BSM particles ALPs~~ to interact with electrons and ~~$F_{\text{EBSM EALP}}$~~ is the fluence of ~~BSM particles with energy E_{BSM} ALPs with energy E_{ALP}~~ . It can be seen in Fig. 2 of [23] that the efficiency of our detector, ϵ for the ER channel is $\sim 89\%$ for recoil energies greater than 5 keV_{ee}. The number of electrons available per atom when the recoil energy is greater than 34.6 (5.5) keV_{ee} is 54 (52) electrons. Given that we do not see any background event, we obtain the upper limit on the product of coincident fluence and cross section: ~~$\sigma_{\text{BSM}} F_{\text{EBSM}} < 10^{-29} \sigma_{\text{ALP}} F_{\text{EALP}} < 10^{-29} \text{ cm}^2/\text{cm}^2$~~ for ~~$E_{\text{BSM EALP}}$~~ in [5.5–210] keV_{ee}.

We have thus provided the first search for particle signals in the keV energy region associated with GWs. Despite the 1.3 tonnes fiducial mass of XENON1T, various ~~recoil channels analysis modes~~, energy regions down to 1 keV_{ee}, no particle signals were found within ± 500 s time windows around GW signals. As a result, we extended

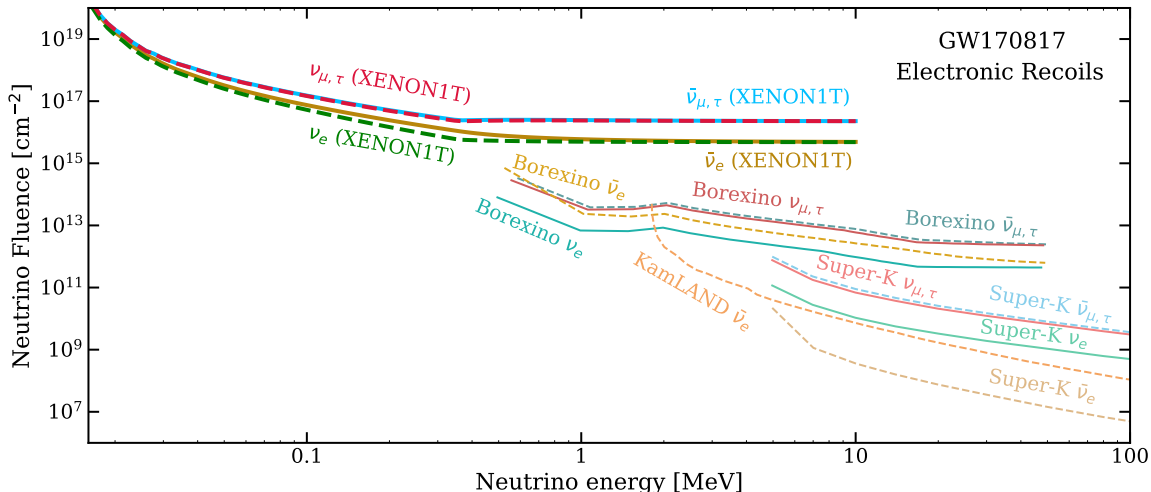


FIG. 3. The upper limit on the fluence (time-integrated flux) of coincident neutrinos emitted in GW170817, the merger of neutron stars using the electronic recoil signals in XENON1T, obtained for electron neutrinos (ν_e , dashed green), electron antineutrinos ($\bar{\nu}_e$, solid gold), muon or tau neutrinos ($\nu_{\mu,\tau}$, dashed red), and muon or tau antineutrinos ($\bar{\nu}_{\mu,\tau}$, solid cyan). We also show the limits obtained by Borexino [10], KamLAND [9], and Super-K [7] for comparison.

the upper limits on the coincident fluence down to neutrino energy of 17 keV. Further, we put an upper limit on the product of coincident fluence and cross section of ~~general BSM particles~~ ALPs interacting with atomic electrons. These limits can help constrain models involving neutrinos or other BSM particles in binary star mergers. As gravitational wave detectors will continue taking data during the operation of the current and next generation of multiple tonnes xenon detectors [38–42], future GWs from nearby mergers will further improve these constraints.

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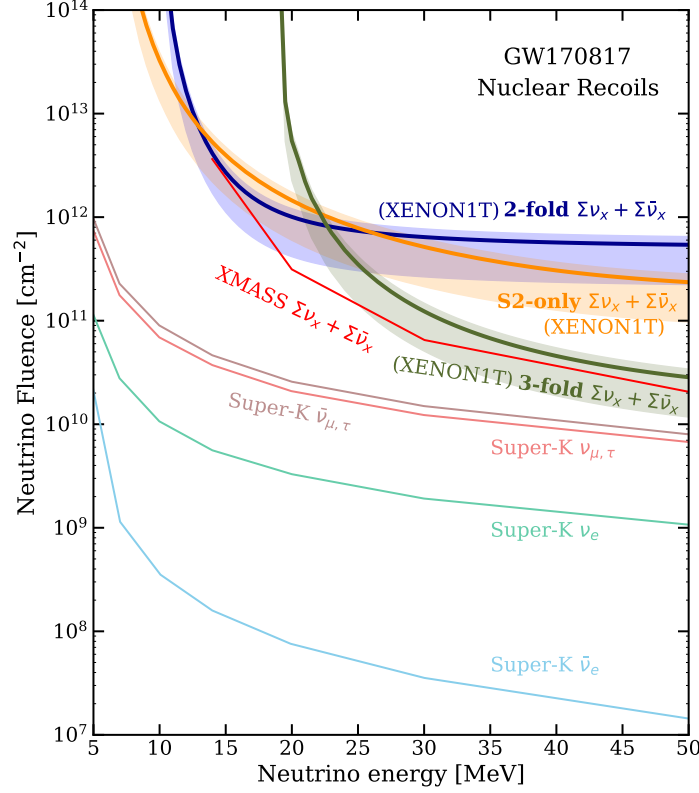


FIG. 4. The upper limit on the fluence (time-integrated flux) of coincident neutrinos emitted in GW170817, the merger of neutron stars using the nuclear recoil signals in XENON1T, placed on the sum of the fluence of all flavors of neutrinos and antineutrinos. **NR-3-fold** (olive-green), **CEvNS-2-fold** (blue), and **S2-only** (orange) refer to different **ways-modes** in which the data was analyzed in **the** search for nuclear recoils, as described in Sec. II. We also show the limits obtained by XMASS [8] and Super-K [7] for comparison.

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- [1] B. P. Abbott et al. (LIGO Scientific, Virgo), Phys. Rev. Lett. **116**, 061102 (2016), 1602.03837.
 - [2] R. Abbott et al. (LIGO Scientific, VIRGO, KAGRA) (2023), 2302.03676.
 - [3] B. P. Abbott et al. (LIGO Scientific, Virgo), Phys. Rev. Lett. **119**, 161101 (2017), 1710.05832.
 - [4] B. P. Abbott et al. (LIGO Scientific, Virgo, Fermi GBM, INTEGRAL, IceCube, AstroSat Cadmium Zinc Telluride Imager Team, IPN, Insight-Hxmt, ANTARES, Swift, AGILE Team, 1M2H Team, Dark Energy Camera GW-EM, DES, DLT40, GRAWITA, Fermi-LAT, ATCA, ASKAP, Las Cumbres Observatory Group, OzGrav, DWF (Deeper Wider Faster Program), AST3, CAAS-TRO, VINROUGE, MASTER, J-GEM, GROWTH, JAGWAR, CaltechNRAO, TTU-NRAO, NuSTAR, Pan-STARRS, MAXI Team, TZAC Consortium, KU, Nordic Optical Telescope, ePESSTO, GROND, Texas Tech University, SALT Group, TOROS, BOOTES, MWA, CALET, IKI-GW Follow-up, H.E.S.S., LOFAR, LWA, HAWC, Pierre Auger, ALMA, Euro VLBI Team, Pi of Sky, Chandra Team at McGill University, DFN, ATLAS Telescopes, High Time Resolution Universe Survey, RIMAS, RATIR, SKA South Africa/MeerKAT), Astrophys. J. Lett. **848**, L12 (2017), 1710.05833.
 - [5] E. Santos (Pierre Auger), J. Phys. Conf. Ser. **1181**, 012060 (2019).
 - [6] R. Abbasi et al. (IceCube) (2023), 2303.15970.
 - [7] Y. Hayato et al. (Super-Kamiokande), Astrophys. J. Lett. **857**, L4 (2018), 1802.04379.
 - [8] K. Abe et al. (XMASS), Astropart. Phys. **129**, 102568 (2021), 2007.16046.
 - [9] S. Abe et al. (KamLAND), Astrophys. J. **909**, 116 (2021), 2012.12053.
 - [10] D. Basilico et al. (BOREXINO), Eur. Phys. J. C **83**, 538 (2023), 2303.13876.

- [11] A. Albert et al. (ANTARES), JCAP **04**, 004 (2023), 2302.07723.
- [12] M. A. Acero et al. (NOvA), Phys. Rev. D **104**, 063024 (2021), 2106.06035.
- [13] E. Aprile et al. (XENON), Eur. Phys. J. C **77**, 881 (2017), 1708.07051.
- [14] S. P. Harris, J.-F. Fortin, K. Sinha, and M. G. Alford, JCAP **07**, 023 (2020), 2003.09768.
- [15] D. F. G. Fiorillo and F. Iocco, Phys. Rev. D **105**, 123007 (2022), 2109.10364.
- [16] M. U. Frensel, Ph.D. thesis, Basel U. (main) (2017).
- [17] M. Frensel, M.-R. Wu, C. Volpe, and A. Perego, Phys. Rev. D **95**, 023011 (2017), 1607.05938.
- [18] J.-S. Wang, F.-K. Peng, K. Wu, and Z.-G. Dai, Astrophys. J. **868**, 19 (2018), 1810.00170.
- [19] G. M. Fuller, A. Kusenko, D. Radice, and V. Takhistov, Phys. Rev. Lett. **122**, 121101 (2019), 1811.00133.
- [20] M. Abdullah et al. (2022), 2203.07361.
- [21] E. Aprile et al. (XENON), Phys. Rev. Lett. **119**, 181301 (2017), 1705.06655.
- [22] E. Aprile et al. (XENON), Phys. Rev. Lett. **121**, 111302 (2018), 1805.12562.
- [23] E. Aprile et al. (XENON), Phys. Rev. D **102**, 072004 (2020), 2006.09721.
- [24] E. Aprile et al. (XENON), Phys. Rev. Lett. **126**, 091301 (2021), 2012.02846.
- [25] E. Aprile et al. (XENON), Phys. Rev. Lett. **123**, 251801 (2019), 1907.11485.
- [26] B. P. Abbott et al. (LIGO Scientific, Virgo), Phys. Rev. X **9**, 031040 (2019), 1811.12907.
- [27] M. Cusinato, F. M. Guercilena, A. Perego, D. Logoteta, D. Radice, S. Bernuzzi, and S. Ansoldi (2021), 2111.13005.
- [28] C. Giunti and A. Studenikin, Phys. Atom. Nucl. **72**, 2089 (2009), 0812.3646.
- [29] P. Vogel and J. Engel, Phys. Rev. D **39**, 3378 (1989).
- [30] M. Winter, *Xenon: Properties of free atoms*, URL <https://www.webelements.com/xenon/atoms.html>.
- [31] B. Baret et al., Astropart. Phys. **35**, 1 (2011), 1101.4669.
- [32] E. Aprile et al. (XENON), Phys. Rev. D **100**, 052014 (2019), 1906.04717.
- [33] E. Shockley, *Study of Excess Electronic Recoil Events in Xenon1T* (2021).
- [34] A. E. Bayer and U. Seljak, JCAP **10**, 009 (2020), 2007.13821.
- [35] Z. Šidák, Journal of the American Statistical Association **62**, 626 (1967).
- [36] G. J. Feldman and R. D. Cousins, Phys. Rev. D **57**, 3873 (1998), physics/9711021.
- [37] A. Derevianko, V. A. Dzuba, V. V. Flambaum, and M. Pospelov, Phys. Rev. D **82**, 065006 (2010), 1007.1833.
- [38] E. Aprile et al. (XENON) (2023), 2303.14729.
- [39] J. Aalbers et al. (LZ) (2022), 2207.03764.
- [40] J. Liu (PandaX), Int. J. Mod. Phys. D **31**, 2230007 (2022).
- [41] J. Aalbers et al., J. Phys. G **50**, 013001 (2023), 2203.02309.
- [42] XLZD, *Dark matter detection consortium*, URL <https://xlzd.org/>.